SAVE MORGAN VALLEY
WEBSITE INFORMATION PACKAGE
Morgan Valley Energy Center (MVEC)

PROJECT OVERVIEW

The proposed Morgan Valley Energy Center is a 720-megawatt natural gas-fired power plant proposed by Interstate Power & Light Company (Alliant Energy) in Linn County, Iowa near Fairfax and Atkins.

According to Interstate Power & Light Company's "PSD Air Construction Permit Application (80% Complete)," dated April 23, 2026, the proposed project would consist of:

- Three simple-cycle natural gas-fired combustion turbines
- Four natural gas-fired hot water boilers
- One emergency diesel generator
- One emergency diesel fire pump
- Haul road traffic fugitives
- Natural gas piping components
- Sulfur hexafluoride (SF_6) containing circuit breakers
- Additional auxiliary equipment

The application states the project would be a "Prevention of Significant Deterioration (PSD) major source" under federal air permitting regulations.

PROJECTED ANNUAL EMISSIONS

According to Interstate Power & Light Company's "PSD Air Construction Permit Application (80% Complete)," dated April 23, 2026, the proposed facility could emit approximately:

GREENHOUSE GASES (CO_2e)
1,516,388 tons per year

- Greenhouse gases (CO_2e) are regulated under federal greenhouse gas permitting requirements for major sources.
- The application identifies the project as a major greenhouse gas source.

CARBON MONOXIDE (CO)
1,152 tons per year

• Carbon monoxide (CO) is a regulated air pollutant produced during combustion.

NITROGEN OXIDES (NOx)
319 tons per year

• NOx contributes to ozone formation and regional smog.
• NOx emissions can also contribute to particulate formation.

VOLATILE ORGANIC COMPOUNDS (VOC)
105 tons per year

• VOCs contribute to ground-level ozone formation.

PARTICULATE MATTER (PM / PM10 / PM2.5)
61.6 tons per year

• PM2.5 refers to extremely small airborne particles capable of entering deep into the lungs.
• Fine particulate matter (PM2.5) is regulated under federal air quality standards due to potential respiratory and cardiovascular health concerns.

SULFUR DIOXIDE (SO2)
18 tons per year

• SO2 is a regulated air pollutant associated with industrial combustion processes.

SULFURIC ACID MIST (H2SO4)
2.7 tons per year

• Sulfuric acid mist is a regulated air pollutant associated with industrial combustion processes.

The application states that multiple pollutants exceed federal PSD significance thresholds and therefore trigger PSD review.

FACTS FROM THE 80% COMPLETE PSD AIR CONSTRUCTION PERMIT APPLICATION

The application states:

“The Project is a Prevention of Significant Deterioration (PSD) major source…”

The application also states:

“Potential emissions from the Project are shown in Table 1-1…”

The application identifies multiple pollutants exceeding federal PSD significance thresholds, including:

- NOx
- Carbon monoxide
- VOCs
- PM / PM10 / PM2.5
- Greenhouse gases

The application discusses multiple air pollution control technologies and evaluates whether they will be used.

POLLUTION CONTROL TECHNOLOGIES DISCUSSED IN THE APPLICATION

SELECTIVE CATALYTIC REDUCTION (SCR)

The application identifies SCR (Selective Catalytic Reduction) as a technically feasible NOx control technology and discusses its evaluation during the BACT analysis.

The application ultimately proposes Dry Low-NOx (DLN) burners as the selected NOx control technology rather than SCR.

The application’s BACT analysis includes discussion of economic impacts when evaluating control technologies.

OXIDATION CATALYSTS

The application also evaluates oxidation catalyst technology capable of reducing carbon monoxide and VOC emissions.

The proposed project instead relies primarily on combustion controls and good combustion practices for those emissions.

CARBON CAPTURE AND SEQUESTRATION

The application discusses carbon capture and sequestration technologies as part of its greenhouse gas analysis.

The project does not propose carbon capture as part of the facility.

SITE-SPECIFIC WEATHER MONITORING

Nearby residents requested permission to place temporary weather monitoring equipment near the proposed project site in order to better understand localized atmospheric conditions.

Alliant Energy declined the request to place independent monitoring equipment on the proposed project site.

Residents were informed that air quality monitoring and environmental review associated with the project are being conducted through the formal regulatory process overseen by Linn County Public Health and the Iowa Department of Natural Resources.

The PSD modeling protocol references meteorological data, ambient monitoring, background concentrations, and air dispersion modeling methodology.

Residents continue raising questions regarding whether site-specific meteorological and baseline air-quality monitoring will be required for the project.

DIRECT QUOTES — ALLIANT ENERGY CORRESPONDENCE

“Projects of this nature often advance through parallel review processes at the state and local levels. This allows technical planning, coordination, and regulatory review to move forward while public awareness and engagement continue.”

— Mayuri Farlinger, President, Interstate Power & Light Company

“We recognize that safety, especially for your children and family, is your primary concern…”

— Ben Rogers, Alliant Energy

“This is not a public forum for Alliant Energy to answer questions on LCPH’s permitting processes.”

— Ben Rogers, Alliant Energy

“As we receive additional information and continue finalizing our air permit application with LCPH, we will look for opportunities to share updates and help keep the community informed.”

— Taylor Adams, Alliant Energy

DIRECT QUOTES — RESIDENT QUESTIONS

“What specific health studies does Alliant Energy have demonstrating that families living approximately one-half mile from a natural gas-fired power plant of this size and magnitude can be expected to remain safe and not experience adverse health impacts?”

“Why can’t the families living closest to this site get clear answers about the long-term health impacts to their children and households?”

“Why is site-specific weather monitoring not being required for a project of this scale?”

“How is it appropriate for the company profiting from this facility to decide, on cost grounds, against control technologies that would substantially reduce emissions, while being unable to show the affected public a single study that nearby families will remain safe?”

DIRECT QUOTES — 80% COMPLETE PSD AIR CONSTRUCTION PERMIT APPLICATION

“The Project is a Prevention of Significant Deterioration (PSD) major source…”

“Potential emissions from the Project are shown in Table 1-1…”

“State-of-the-art pollution control equipment has been selected as BACT for the Project.”

“Emissions of NOx from the combustion turbines will be controlled by dry low-NOx (DLN) burners.”

FREQUENTLY ASKED QUESTIONS (FAQ)

WHAT IS THE MORGAN VALLEY ENERGY CENTER?

The Morgan Valley Energy Center is a proposed 720-megawatt natural gas-fired power plant proposed by Interstate Power & Light Company (Alliant Energy) near Fairfax and Atkins in Linn County, Iowa.

WHAT DOES “PSD MAJOR SOURCE” MEAN?

PSD stands for Prevention of Significant Deterioration, a major federal air permitting program under the Clean Air Act.

A PSD major source is a facility projected to emit pollutants above specific federal significance thresholds.

WHAT IS PM2.5?

PM2.5 refers to extremely small airborne particles that are small enough to enter deep into the lungs.

WHAT IS NOx?

NOx stands for nitrogen oxides, pollutants produced during combustion that contribute to ozone formation and smog.

WHAT IS SCR?

SCR stands for Selective Catalytic Reduction, an emissions-control technology commonly used to reduce NOx emissions from industrial combustion sources.

WHAT IS DLN?

DLN stands for Dry Low-NOx burners, a combustion-based NOx reduction technology proposed for the Morgan Valley project.

WILL THE PROJECT USE SITE-SPECIFIC WEATHER MONITORING?

Residents requested site-specific weather monitoring near the proposed site. Questions remain regarding whether direct, long-term localized monitoring will ultimately be required.

HAS THE PROJECT BEEN APPROVED?

No. The project remains subject to multiple state and local review, permitting, zoning, and approval processes.

WHY ARE RESIDENTS ASKING QUESTIONS ABOUT AIR QUALITY?

Residents have raised concerns regarding:
• projected emissions quantities
• proximity to homes
• schools and hospitals inside the impact area
• lack of site-specific monitoring
• long-term industrial growth
• and pollution-control technology decisions

PRIMARY SOURCE DOCUMENTS

Recommended website downloads/links:

- “PSD Air Construction Permit Application (80% Complete)”
- PSD Class II Air Dispersion Modeling Protocol
- Iowa Utilities Commission docket filings
- Public meeting notices
- Public comments and correspondence
- Linn County Public Health materials
- DNR correspondence and permitting information

The public is encouraged to review original source documents directly.